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To cite this article: A. Uzunov, E. W. Brascamp & R. Büchler (2017) The Basic Concept of Honey Bee Breeding Programs, Bee World, 94:3, 84-87, DOI: [10.1080/0005772X.2017.1345427](https://doi.org/10.1080/0005772X.2017.1345427)

To link to this article: <https://doi.org/10.1080/0005772X.2017.1345427>



Published online: 11 Aug 2017.



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The Basic Concept of Honey Bee Breeding Programs

A. Uzunov, E. W. Brascamp and R. Büchler

Selective honey bee breeding is a phenomenon that fascinates beekeepers around the world. They often regard it as one of the most enigmatic and complex aspects of beekeeping. Indeed, according to our experiences participating in many international projects, both beekeepers and bee experts without a background in plant or animal breeding often have trouble correctly interpreting and conceptually visualizing the breeding process. These difficulties arise partly because of the complex reproductive biology of honey bees, where queens mate with a multitude of drones. Fundamentally the greatest struggle for people to understand is how selection of animals with preferred characteristics in one generation leads to improved progeny in the next.

The leading misconception regarding honey bee breeding is confusing breeding with the simple rearing and multiplication of queens, where individual queens are evaluated predominantly by their egg laying ability and body size. Those two markers of queen quality (fecundity and size) are certainly important for the propagation of queens, but selective breeding requires more than propagation. Selective breeding implies the intentional selection for genetic improvement of the population as a whole, with every new generation improved compared to the previous one, ideally for all traits of interest.

To achieve such improved selection, queens and colonies in each generation must be chosen that exhibit the desirable properties the breeder wishes to propagate, using these as parents for the next generation so that on average the next generation is expected to be better than the previous one. Breeding success for a certain honey bee population is thus the cumulative outcome of many actions, such as performance testing on the colony, the selection and mating of individuals.

Our aim with this paper is to help demystify conventional honey bee breeding programs and highlight the typical features of successful programs, based on the breeding experiences in Central Europe. We focus on conceptual aspects beekeepers and breeders can control without discussing the underlying theories based upon population and quantitative genetics.

A Breeding Program and Its Elements

A breeding program represents a set of systematically planned and implemented activities aimed at the sustained genetic improvement of a honey bee population (Brascamp, 2014; Tiesler et al., 2016). Thus, by continuous implementation of this selection program it is expected that the colonies in the next generation will express improved behavior concerning

targeted traits (for instance: gentleness, calmness on the comb, reduced swarming), enhanced production (honey, pollen, wax, royal jelly) and vitality (resistance to diseases and pests, prolonged life expectancy, etc.).

A breeding program should include explicit breeding objectives, performance testing to evaluate the desired characteristics, estimation of the breeding values, selection, mating, multiplication of the improved genetic stock, and evaluation (Figure 1).

The first and very important step is to define the *breeding objective*. Thus, based on the economic importance, scientific evidence and practical experience, breeders must decide which traits they intend to improve, and what is the relative importance of improving the different traits.

Generally, the preferred traits in selection for honey bees involve improving honey yield, gentleness, decreasing the swarming tendency and increasing *Varroa* resistance. Sometimes traits such as coloration, which are typical for a particular population, can be included in the selection process. In principle all traits of importance should be included, because selection for too limited a set of traits may lead to deterioration of other relevant traits. For example selecting only for *Varroa* resistance with no regard for the qualities that make for productive colonies can result in queens that are not practical for commercial beekeeping.

The German breeding program for resistance to *Varroa* (AGT) is an example where the focus is on *Varroa* resistance (Table 1), but other traits are also taken into account. The decision regarding which traits to include to achieve the breeding objective depends mainly on the interests of a particular group of breeders. Objectives such as the conservation of an isolated population or specific research goals can be relevant too.

Since breeding success is accumulated genetic improvement over time the

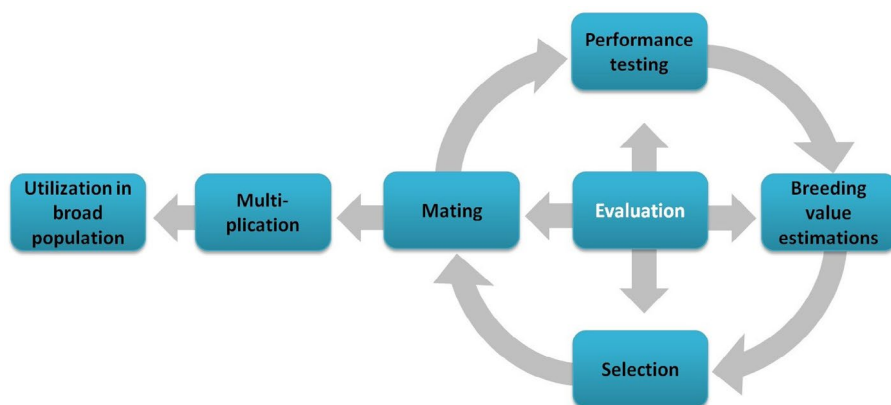


Figure 1. Elements of a breeding program and sequence of actions. A continual process of selecting towards the breeding objective. Mating is carried out to produce a next generation for performance testing, but also for multiplication of the improved material to be disseminated and utilized in the broad population.

Table 1. The weighting of desirable traits in the AGT breeding program.

Trait	Honey yield	Gentleness	Calmness	Swarming behavior	Varroa index*
Weighting of traits (%) in the total breeding value	15	15	15	15	40

*A combination of natural mite fall in spring, mite infestation in late summer and hygienic behavior.

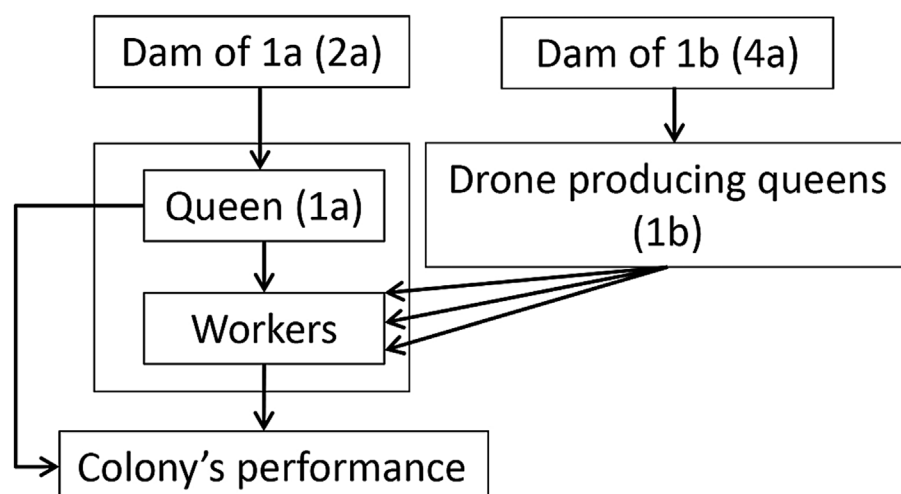


Figure 2. Effects of the workers and queen on the colony's performance.

breeding objective ideally should be consistent from year to year and not subject to frequent drastic changes. Thus, before the objective is defined one should foresee the needs and demands that might occur in the future, so that the selection has long-term relevance. Finally, the breeding objective should take into account the way bees are managed and how beekeeping operations are organized if the goal is improved stock for beekeepers.

Performance testing requires a standardized methodology for the phenotypic assessment of traits and accurate data recording (Büchler et al., 2013; Ruttner, 1972; Uzunov et al., 2015). While in farm

animals performance is generally measured on individuals (milk production in dairy cows, egg laying rate in hens), in honey bees traits are usually observed on the colony level. This is illustrated in Figure 2.

On the one hand the workers contribute to colony's performance, known as the worker effect, but the queen (1a) also has a direct effect (queen effect) on the colony's performance. Of course she also contributes indirectly, because she transmits her breeding value for the worker effect to the workers. As an example, honey yield is affected by the ability of workers to collect nectar, but also by the egg laying capacity

and pheromone production of the queen. Figure 2 also illustrates that apart from the queen (1a) that is the mother of the colony, the drone-producing queens (1b) also contribute 50% to the breeding value of the workers. Since drones are haploid and have no father, it is often said that the paternal contributor to the breeding value of the workers in practice is 4a, the so-called father colony.

A written guideline for performance testing (protocol), along with practice and training of the bee breeders in the proposed methods (breeders' capacity building) are essential so that test colonies can be evaluated objectively and accurately across apiaries and operations. Before engaging in a breeding program and queen evaluation, a number of preparatory activities should be conducted. These include establishing test apiaries and colonies, as well as organizing a queen exchange among the different test apiaries.¹

The next step is the estimation of breeding values. By definition the breeding value of an individual is the expected performance of the offspring when the individual is mated to an average mate and the offspring is performing in an average environment. Usually these complicated calculations of the breeding value estimation are conducted by the facility that manages the database of performance testing results.

A best case study of such a system is represented by BEEBREED (www.beebreed.eu) where via an online web-based platform the performance testing data are stored, breeding values are estimated and subsequently published. In BEEBREED the estimated breeding value of an individual takes into account the colony's individual performance as well as the performances of other colonies in the same environment (test apiary) and performances of colonies that are ancestors or other relatives (Figure 3). Comparison of colony's performance with that of colonies at the same test apiary takes into account differences caused by beekeeping techniques applied, weather, food sources etc. For more details, see Bienefeld et al. (2007), Büchler et al. (2013), Brascamp et al. (2016) and Tiesler et al. (2016).

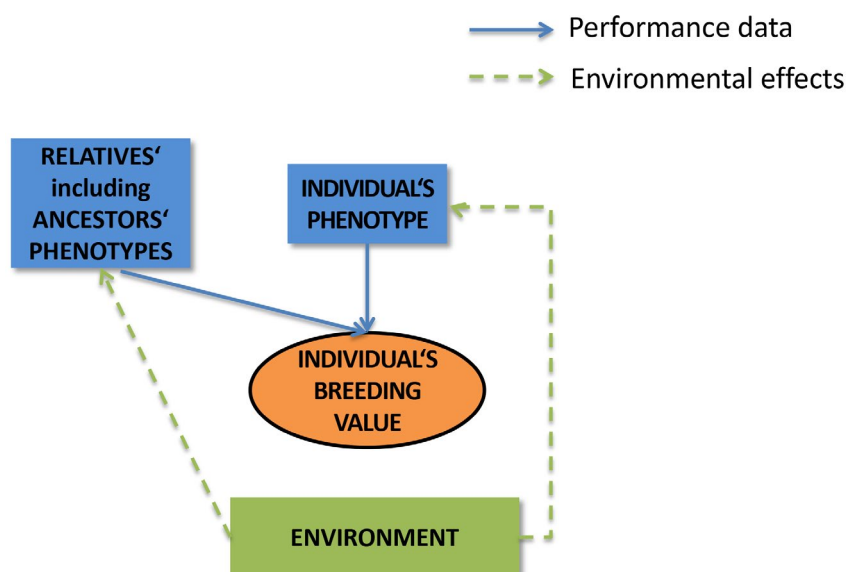


Figure 3. Simplified model of the effects contributing to the estimation of breeding values according to BLUP Animal Model.

To obtain reliable breeding values, it is important that daughters of each breeder queen are tested at several test apiaries.

The selection of queens based upon ranking for estimated breeding values is a further step in the breeding program. Usually individual breeders not only select based upon this criterion but include additional observations such as overwintering ability, colony strength, etc. Thus queens are ranked and selected as mothers for the next generation of queens as well as queens to produce drone producing queens. Often individual breeders decide what queens they will use to produce the next generation, while an association is often responsible for selecting the queen that will be the mother of the queens that head the multiple drone colonies at a mating station, under the assumption that such a mating station is not used by one single breeder. Via instrumental insemination or private isolated mating yards, an individual breeder can control which queen stock serves as the paternal ancestor of the next generation.

In honey bees *mating* deserves specific attention due to the reproductive behavior of the species. Special measures are required to control mating, for example through the use of mating stations (isolated mainland or island locations) or instrumental insemination. According to theory the breeding value of a progeny equals half the sum of the breeding values of the parents. As a consequence, parents can be selected independently and mated at random when it concerns the average breeding value of the next generation.

However, the desire of individual breeders to compensate for a weakness in a selected queen with the strength of the mate advocates for the combined selection of a queen and her mates and not just the individual parents. For example a queen that has excellent *Varroa* resistance and honey production, but a higher swarm tendency might be mated with drones that originated from a queen that exhibited a very low swarm drive. Another benefit of selecting both partners is that potential inbreeding of workers can be taken into account.

Genetic improvement is the ultimate goal of a breeding program and involves eventual propagation of the improved stock in the general honey bee population (Büchler et al., 2010; Rinderer et al., 2010). The beekeeping community can incorporate improved stock on the maternal side in the form of grafted larvae, queens cells, virgin or mated queens. Through the use

of drone semen in instrumental insemination or the mating stations where selected queens produce drones that will mate the virgin queens, beekeepers can incorporate improved genetic stock on the paternal side, helping to improve the quality of the bee population.

We feel that the topic of propagation of improved stock does not receive the attention it deserves. Activities concerning this goal, like transparent communication, marketing and efficient delivery systems, should be well planned and implemented and require concerted actions of the beekeeping community at large.

Finally, a program should systematically be subjected to evaluation. This should be recognized as a distinct element. The first step is to compare the predicted and realized selection response or “does the program meet its promises?”. In the evaluation, all elements and steps of the breeding program should be scrutinized and any obstacles should be identified and removed.

Here issues like the accuracy of the methodology for data gathering, the recording of pedigrees, and the accuracy of breeding value estimations should be evaluated, as well as re-evaluating strategic goals and delineating future expectations. In addition, breeding programs should consider if the environments in which the daughter queens of the breeding program perform are sufficiently uniform, such that the colonies produced from the better line in one environment are consistently the better ones in another environment.

Types of Breeding Programs

In the previous paragraphs we described a breeding program based strongly on the results of BEEBREED and the beekeeping community in Europe. However, the reasons to keep, and possibly improve honey bees, differ among bee breeders around the world.

There is a wide spectrum of breeding programs. To illustrate this we list the most common types of breeding programs.

- **Commercial**, aimed to improve the overall performance of the honey bees from the population of interest, based upon the assessment of various traits. Occasionally, the number of traits is limited to 3 or 4. However, the main drive and objective of these commercial

breeding programs is improvement in commercially important traits (more honey, less defensive bees, reduced swarming tendency, etc.). This type of selection is most common and considered by us as the most sustainable.

- **Conservation**, aimed at the maintenance of endangered honey bee populations. The ultimate goal is maintenance or enlargement of the population. Genetic improvement of such populations is a useful tool in the context of “conservation by utilization”, as conservation by utilization is considered a preferred mechanism to conserve subspecies or populations. Along with commonly recognized traits, relevant on the regional or local scale, morphological characters and molecular markers are frequently the basis for decision-making and selection, the latter two being used to ensure that the population is not mixed with other subspecies.
- **Research** breeding programs can be initiated for studying certain traits (effects of the genes, identification of markers, etc.) of scientific interest as well as analyzing the effects of hybridization or inbreeding, assessing the adaptive ability of populations, resistance to diseases, genotype by environment interactions, etc. Generally these breeding programs are short-term and under the responsibility of research institutes or other academic institutions.

Whatever type of breeding program is initialized there is always the question of the genetic origin of the population under selection. In general there are two main approaches. Recently a trend has emerged for sustainable conservation, a popular breeding approach to improve and conserve the native or locally adapted honey bee populations or subspecies. The basic philosophy behind this is to reduce importations and instead utilize and improve the local populations in comparison to the non-local ones.

Alternatively, often breeders start a breeding program building upon breeds like Ligustica, Carnica or Buckfast (which can be considered a breed of recent origin) with assumed general usefulness across environments. From a conservation point of view the disadvantage of this is that the mixing of local populations and these general purpose populations, when kept in the same area, is difficult to prevent.

Whatever approach is chosen the main issue during the initiation of the breeding program is appropriate selection of the

first genotypes. Here the best advice is to choose the best available queens, preferably based on a breeder's records, taking care that the founding queens represent sufficient genetic diversity.

Conducting the Breeding Program

Breeders, regional groups, scientists, national and local authorities are the main partners of a breeding program. The organization of regular meetings is important for better coordination of the activities as well as for exchange of ideas and overcoming the challenges of implementation. A close cooperation with scientists can significantly help the program achieve its breeding objective, by aiding in the development of standardized methodology and introducing new approaches and models of breeding.

Concerning the establishment of controlled mating locations, the cooperation with local authorities may be important. If we take in to consideration the above mentioned elements, the overall success and sustainability of a breeding program significantly depends on the collaboration, communication, transparency and

exchange of ideas among all stakeholders within and around the breeding program.

Acknowledgment

We would like to thank Dr. Bjorn Dahle, Dr. Sreten Andonov, Dr. Eliza Cauia, Egoitz Galarza, Borce Pavlov, Dr. Andreas Hoppe and Dirk Ahrens-Lagast for their valuable comments and suggestions for improvement of the readability of the article and integration of experience from various breeding practices across Europe.

Note

1. Location or apiary where testing of the colonies is performed.

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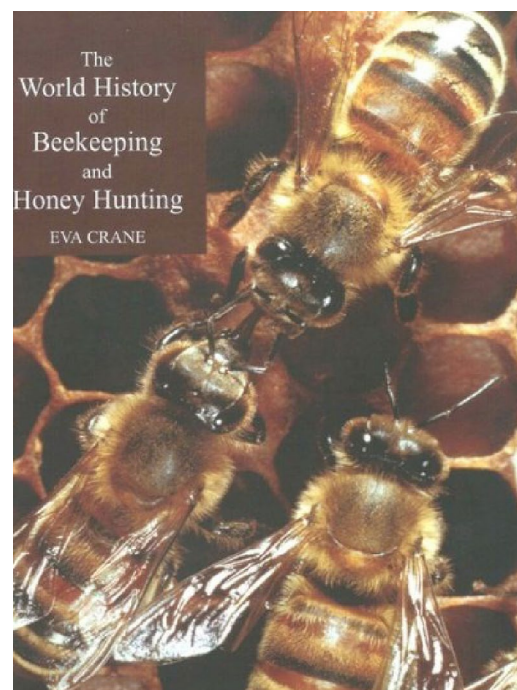
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Eva Crane's classic "World history..."

This comprehensive book is the first to explore in detail the world history of man's use of bees from prehistoric times to the present day. It gathers together a vast amount of information in an eminently readable text. From archaeological evidence about bee hives in Ancient Egypt to the Maya people of Mesoamerica who kept stingless bees, Dr Crane recognises the variations in methods and yet there are some uncanny similarities. It wasn't until the 1600 that beekeeping techniques started to change in Europe, which culminated in 1851 with Langstroth's moveable frame hives being produced in the USA. The subsequent changes in beekeeping, based on scientific knowledge, brings the history up to date. There are over 500 illustrations and extensive bibliographies and appendices.



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